



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2017
&ANSI/NCSL Z540-1-1994

CIH EQUIPMENT COMPANY, INC.
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CALIBRATION

Valid To: August 31, 2023

Certificate Number: 3035.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations^{1,4}:

I. Acoustical Quantities

Parameter/Equipment	Range	CMC ^{2,3} (±)	Comments
Calibrate Acoustical Calibrator –			
Sound Pressure Level	(74 to 114) dB (114 to 124) dB	0.17 % 0.10 %	Acoustical calibrator calibration system
Frequency	31 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.32 % 0.10 % 0.10 %	
Sound Level Meters –			
Electrical Calibration	(78.6 to 95.5) dB (95.5 to 104.6) dB (104.6 to 119.3) dB	1.9 % 1.2 % 1.0 %	Signal generator

II. Fluid Quantities

Parameter/Equipment	Range	CMC ^{2, 3, 5} (±)	Comments
Velocity – Measuring Equipment	(100 to 200) sfpm (201 to 1000) sfpm (1001 to 3000) sfpm (3001 to 5800) sfpm (1200 to 3000) sfpm (3001 to 5000) sfpm (5001 to 7000) sfpm (7001 to 9500) sfpm	6.4 % 1.6 % 1.7 % 1.4 % 0.7 % 1.3 % 1.4 % 1.2 %	Flotek 360 wind tunnel with thermo-anemometer TSI standard WT-4401S wind tunnel with differential pressure standard
ISO Standard Dust Concentration – Measuring Equipment	(5 to 500) mg/m ³	0.03 mg/m ³	Laminar dust chamber system, gravimetric standard
Calibrate Flow Calibrators	(5 to 50) mLPM (51 to 125) mLPM (126 to 250) mLPM (251 to 375) mLPM (376 to 450) mLPM (1 to 7) LPM (8 to 14) LPM (15 to 21) LPM (22 to 30) LPM	2.1 % 0.89 % 0.68 % 0.51 % 0.44 % 0.84 % 0.61 % 0.45 % 0.35 %	Piston Provers – ML800-10 ML800-44
Calibration of Air Sampling Pump – Air Flow	5 ccm to 500 mLPM 50 ccm to 5 LPM (0.3 to 30) LPM	0.27 % 0.35 % 0.30 %	Piston Provers – Definer 220L low flow Definer 220M medium flow Definer 220H high flow
Mass Flow Meters	(0.1 to 1) LPM (1 to 100) LPM (100 to 190) LPM	0.18 LPM 1.7 % 1.3 %	Mass flow meter – TSI

Parameter/Equipment	Range	CMC ^{2, 3, 5} (±)	Comments
Flow (Air)	(400 to 500) CFM (501 to 1200) CFM (1201 to 1800) CFM (1801 to 2600) CFM	1.8 % 1.0 % 0.68 % 0.69 %	Dwyer manometer calibrated using TSI air velocity transducer

III. Magnetic Quantities

Parameter/Equipment	Range	CMC ^{2, 3} (±)	Comments
Gauss Meter – Axial Probe	(1 to 100) G (101 to 1000) G	3.5 % 0.92 %	Standard gauss meter
Flat/Transverse Probe	(7 to 24) G (100 to 10 000) G	3.5 % 0.92 %	

IV. Mechanical

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Pressure – Measuring Equipment	(4 to 150) in·H ₂ O	0.17 in·H ₂ O	Ametek dead weight tester
Force – Measure and/or Measuring Equipment			
Tension	Up to 5 lbf (5 to 10) lbf (10.1 to 100) lbf (100.1 to 200) lbf	0.42 lbf 0.52 lbf 0.80 lbf 0.65 lbf	PCE load cell with universal testing machine
Compression	Up to 5 lbf (5 to 10) lbf (10.1 to 100) lbf (100.1 to 200) lbf	0.46 lbf 0.37 lbf 0.36 lbf 1.4 lbf	

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Torque – Measuring Equipment	(5 to 500) ft·lbf (501 to 1000) ft·lbf	0.73 % 0.86 %	CDI torque calibrator

V. Thermodynamics

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Temperature – Measuring Equipment	(32 to 80) °F	0.57 °F	Rotronic temperature standard
	(81 to 140) °F	0.19 °F	
	(40 to 100) °F	0.08 °F	Vaisala HMT333
	(101 to 140) °F	0.18 °F	
Humidity – Measuring Equipment	(10 to 49) % RH	0.16 % RH	Rotronic humidity standard
	(50 to 90) % RH	0.75 % RH	
	(25 to 50) % RH	0.32 % RH	Vaisala HMT333
	(50 to 95) % RH	0.83 % RH	

¹ This laboratory offers commercial calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ In the statement of CMC, percentages are percentages of the reading/output, unless otherwise noted.

⁴ This scope meets A2LA's *P112 Flexible Scope Policy*.

⁵ The type of instrument or material being calibrated is defined by the parameter. This indicates the laboratory is capable of calibrating instruments that measure or generate the values in the ranges indicated for the listed measurement parameter



Accredited Laboratory

A2LA has accredited

CIH EQUIPMENT COMPANY, INC

Clearwater, FL

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated April 2017).



Presented this 23rd day of September 2021.

A blue ink signature of the Vice President of Accreditation Services.

Vice President, Accreditation Services
For the Accreditation Council
Certificate Number 3035.01
Valid to August 31, 2023

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.